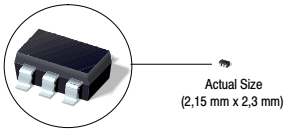


50 mA, 24 V, 3.2 µA Supply Current Low-Dropout Linear Regulator in SC70 Package

FEATURES

- **24-V Maximum Input Voltage**
- **Low 3.2-µA Quiescent Current at 50 mA**
- **Stable With Any Capacitor (≥ 0.47 µF)**
- **50-mA Low-Dropout Regulator**
- **Available in 1.8 V, 1.9 V, 2.3 V, 2.5 V, 3.0 V, 3.3 V, 3.45 V, 5.0 V, and Adjustable (1.2 V to 15 V)**
- **Designed to Support MSP430 Families:**
  - **1.9 V version ensured to be higher than minimum V_{IN} of 1.8 V**
  - **2.3 V version ensured to meet 2.2 V minimum V_{IN} for FLASH on MSP430F2xx**
  - **3.45 V version ensured to be lower than maximum V_{IN} of 3.6 V**
  - **Wide variety of fixed output voltage options to match V_{IN} to the minimum required for desired MSP430 speed**
- **Minimum/Maximum Specified Current Limit**
- **5-Pin SC70/SOT-323 (DCK) Package**
- **-40°C to 125°C Specified Junction Temperature Range**

DESCRIPTION

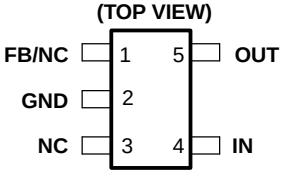
The TPS715xx low-dropout (LDO) voltage regulators offer the benefits of high input voltage, low-dropout voltage, low-power operation, and miniaturized packaging. The devices, which operate over an input range of 2.5 V to 24 V, are stable with any capacitor (≥ 0.47 µF). The low dropout voltage and low quiescent current allow operations at extremely low power levels. Therefore, the devices are ideal for powering battery management ICs. Specifically, since the devices are enabled as soon as the applied voltage reaches the minimum input voltage, the output is quickly available to power continuously operating battery charging ICs.

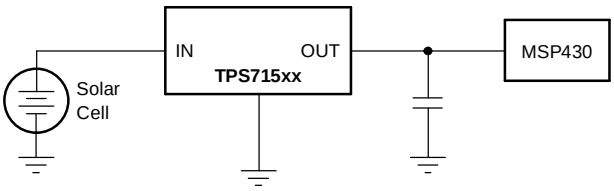
The usual PNP pass transistor has been replaced by a PMOS pass element. Because the PMOS pass element behaves as a low-value resistor, the low dropout voltage, typically 415 mV at 50 mA of load current, is directly proportional to the load current. The low quiescent current (3.2 µA typically) is stable over the entire range of output load current (0 mA to 50 mA).

APPLICATIONS

- **Ultra Low Power Microcontrollers**
- **Cellular/Cordless Handsets**
- **Portable/Battery-Powered Equipment**

DCK PACKAGE
(TOP VIEW)





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This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

ORDERING INFORMATION(1)

PRODUCT	V_{OUT}(2)
TPS715***xxyyyz***	**XX** is nominal output voltage (for example, 28 = 2.8V, 285 = 2.85V, 01 = Adjustable). **YYY** is package designator. **Z** is package quantity.

(1) For the most current package and ordering information, see the Package Option Addendum at the end of this document, or see the TI website at www.ti.com.
(2) Output voltages from 1.2V to 4.5V in 50mV increments are available through the use of innovative factory EEPROM programming; minimum order quantities may apply. Contact factory for details and availability.

ABSOLUTE MAXIMUM RATINGS

over operating temperature range (unless otherwise noted)(1)(2)

	UNIT
V_{IN} range	-0.3 V to 24 V
Peak output current	Internally limited
ESD rating, HBM	2 kV
ESD rating, CDM	500 V
Continuous total power dissipation	See Dissipation Rating Table
Junction temperature range, T_J	-40°C to 150°C
Storage temperature range, T_{stg}	-65°C to 150°C

(1) Stresses beyond those listed under *absolute maximum ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *recommended operating conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) All voltage values are with respect to network ground terminal.

DISSIPATION RATING TABLE

BOARD	PACKAGE	$R_{\theta JC}$°C/W	$R_{\theta JA}$°C/W	DERATING FACTOR ABOVE T_A = 25°C	$T_A \leq$ 25°C POWER RATING	T_A = 70°C POWER RATING	T_A = 85°C POWER RATING
Low-K(1)	DCK	165	395	2.52 mW/°C	250 mW	140 mW	100 mW
High-K(2)	DCK	165	315	3.18 mW/°C	320 mW	175 mW	130 mW

(1) The JEDEC Low-K (1s) board design used to derive this data was a 3 inch x 3 inch, two-layer board with 2 ounce copper traces on top of the board.
(2) The JEDEC High-K (2s2p) board design used to derive this data was a 3 inch x 3 inch, multilayer board with 1 ounce internal power and ground planes and 2 ounce copper traces on top and bottom of the board.

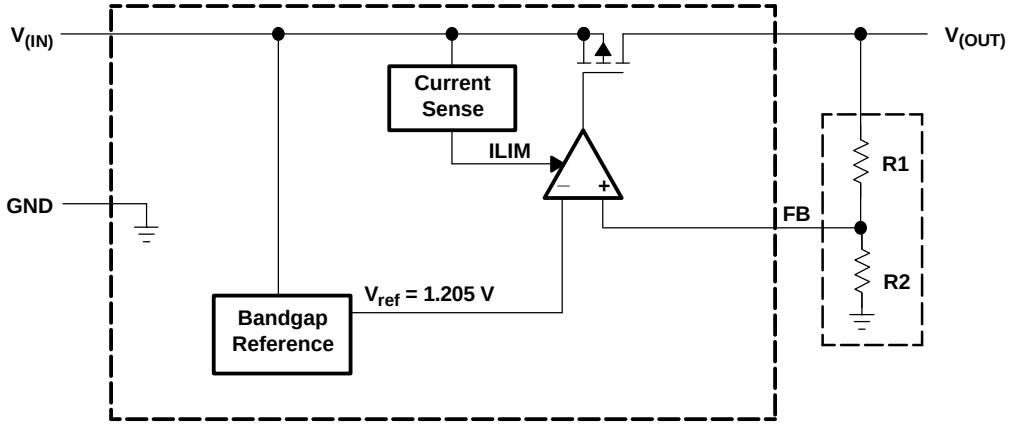
ELECTRICAL CHARACTERISTICS

over operating junction temperature range (T_J = -40°C to 125°C), $V_{IN} = V_{OUT(NOM)}$ + 1 V, I_{OUT} = 1 mA, C_{OUT} = 1 µF unless otherwise noted. Typical values are at T_J = 25°C.

PARAMETER			TEST CONDITIONS	MIN	TYP	MAX	UNIT
Input voltage(1)		V_{IN}	I_O = 10 mA	2.5		24	V
			I_O = 50 mA	3		24	
V_{OUT} voltage range (TPS71501)				1.2		15	V
V_{OUT} accuracy(1)	Over V_{IN}, I_{OUT}, and T		V_{IN} + 1.0 V ≤ V_{IN} ≤ 24 V 100 µA ≤ I_{OUT} ≤ 50 mA	-4.0		+4.0	%
Ground pin current		I_{GND}	0 ≤ I_{OUT} ≤ 50 mA, T_J = -40°C to +85°C		3.2	4.2	µA
			0 mA ≤ I_{OUT} ≤ 50 mA		3.2	4.8	
			0 mA ≤ I_{OUT} ≤ 50 mA, V_{IN} = 24 V			5.8	
Load regulation	$\Delta V_{OUT}/\Delta I_{OUT}$		I_{OUT} = 100 µA to 50 mA		22		mV
Output voltage line regulation (1)	$\Delta V_{OUT}/\Delta V_{IN}$		V_{OUT} + 1 V < V_{IN} ≤ 24 V		20	60	mV
Output noise voltage		V_n	BW = 200 Hz to 100 kHz, C_{OUT} = 10 µF, I_{OUT} = 50 mA		575		µVrms
Output current limit		I_{CL}	V_O = 0 V	125		750	mA
Power-supply ripple rejection		PSRR	f = 100 kHz, C_{OUT} = 10 µF		60		dB
Dropout voltage $V_{IN} = V_{OUT(NOM)}$ - 1 V		V_{DO}	I_{OUT} = 50 mA		415	750	mV

(1) Minimum $V_{IN} = V_{OUT} + V_{DO}$ or the value shown for *Input voltage* in this table, whichever is greater.

FUNCTIONAL BLOCK DIAGRAM—ADJUSTABLE VERSION



FUNCTIONAL BLOCK DIAGRAM—FIXED VERSION

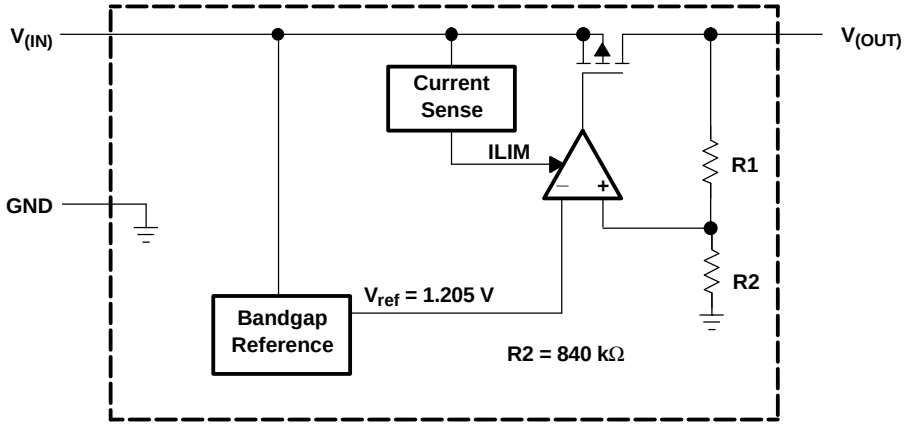


Table 1. Terminal Functions

TERMINAL			DESCRIPTION
NAME	NO.		
	FIXED	ADJ.	
FB		1	Adjustable version. This terminal is used to set the output voltage.
NC	1		No connection
GND	2	2	Ground
NC	3	3	No connection
IN	4	4	Input supply.
OUT	5	5	Output of the regulator, any output capacitor ≥ 0.47 µF can be used for stability.

TYPICAL CHARACTERISTICS

OUTPUT VOLTAGE vs OUTPUT CURRENT

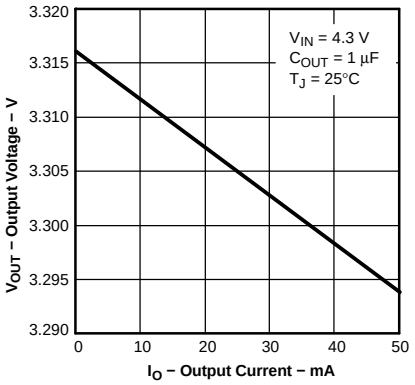


Figure 1.

OUTPUT VOLTAGE vs JUNCTION TEMPERATURE

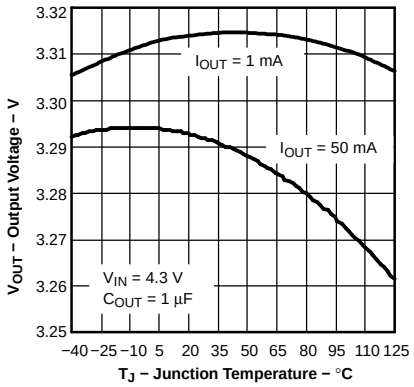


Figure 2.

QUIESCENT CURRENT vs JUNCTION TEMPERATURE

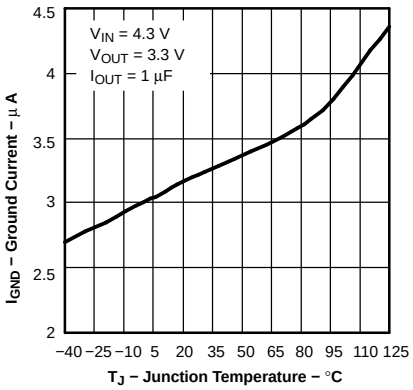


Figure 3.

OUTPUT SPECTRAL NOISE DENSITY vs FREQUENCY

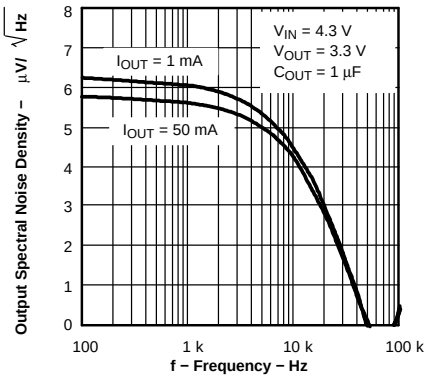


Figure 4.

OUTPUT IMPEDANCE vs FREQUENCY

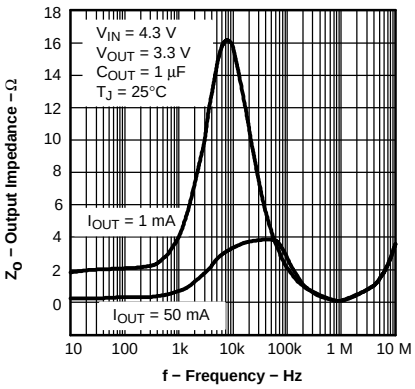


Figure 5.

DROPOUT VOLTAGE vs OUTPUT CURRENT

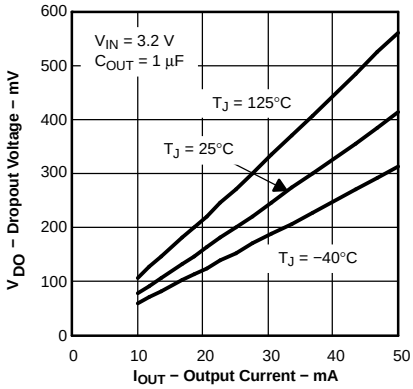


Figure 6.

TPS71501 DROPOUT VOLTAGE vs INPUT VOLTAGE

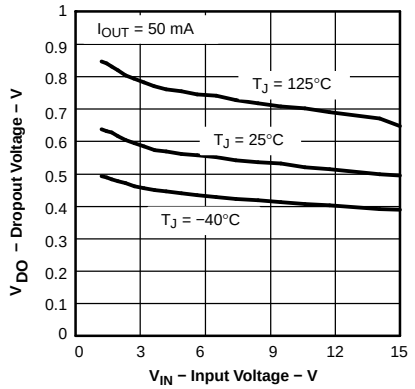


Figure 7.

DROPOUT VOLTAGE vs JUNCTION TEMPERATURE

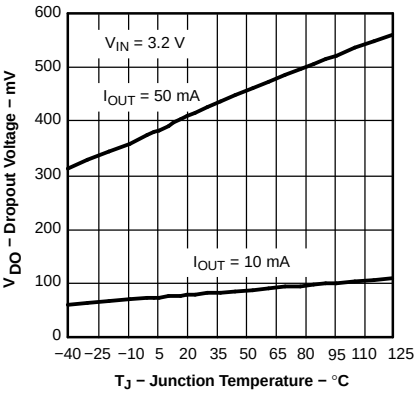


Figure 8.

POWER-SUPPLY RIPPLE REJECTION vs FREQUENCY

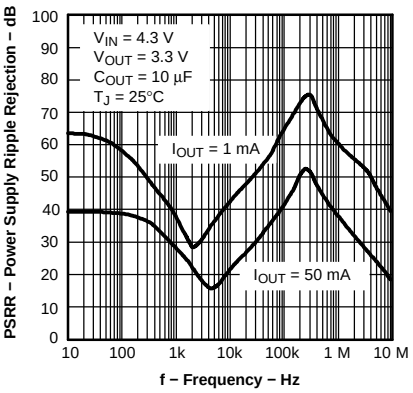


Figure 9.

TYPICAL CHARACTERISTICS (continued)

POWER UP / POWER DOWN

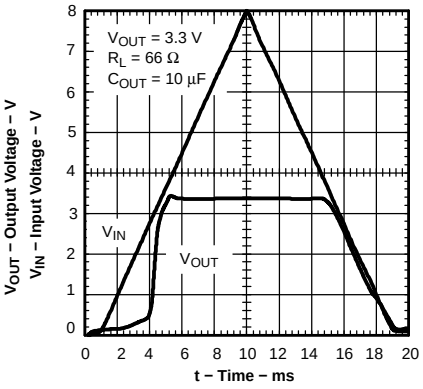


Figure 10.

LINE TRANSIENT RESPONSE

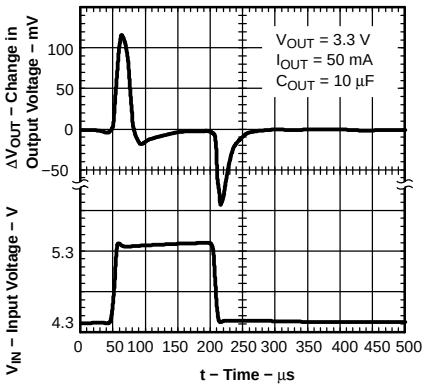


Figure 11.

LOAD TRANSIENT RESPONSE

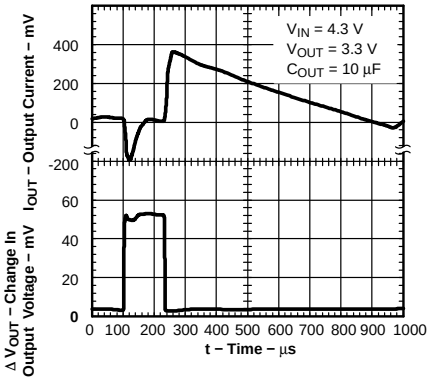


Figure 12.

APPLICATION INFORMATION

The TPS715xx family of LDO regulators has been optimized for ultra-low power applications such as the MSP430 microcontroller. Its ultralow supply current maximizes efficiency at light loads, and its high input voltage range makes it suitable for supplies such as unconditioned solar panels.

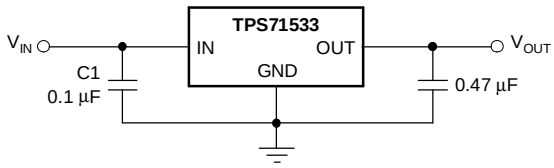


Figure 13. Typical Application Circuit (Fixed Voltage Version)

External Capacitor Requirements

Although not required, a 0.047-µF or larger input bypass capacitor, connected between IN and GND and located close to the device, is recommended to improve transient response and noise rejection of the power supply as a whole. A higher-value input capacitor may be necessary if large, fast-rise-time load transients are anticipated and the device is located several inches from the power source.

The TPS715xx requires an output capacitor connected between OUT and GND to stabilize the internal control loop. Any capacitor (including ceramic and tantalum) ≥ 0.47 µF properly stabilizes this loop.

Power Dissipation and Junction Temperature

To ensure reliable operation, worst-case junction temperature should not exceed 125°C. This restriction limits the power dissipation the regulator can handle in any given application. To ensure the junction temperature is within acceptable limits, calculate the maximum allowable dissipation, $P_{D(max)}$, and the actual dissipation, P_D, which must be less than or equal to $P_{D(max)}$.

The maximum-power-dissipation limit is determined using the following equation:

$$P_{D(max)} = \frac{T_J max - T_A}{R_{\theta JA}} \quad (1)$$

where:

- $T_J$max is the maximum allowable junction temperature.
- $R_{\theta JA}$ is the thermal resistance junction-to-ambient for the package (see the Dissipation Ratings table).
- T_A is the ambient temperature.

The regulator dissipation is calculated using:

$$P_D = (V_{IN} - V_{OUT}) \times I_{OUT} \quad (2)$$

Power dissipation resulting from quiescent current is negligible.

Regulator Protection

The TPS715xx PMOS-pass transistor has a built-in back diode that conducts reverse current when the input voltage drops below the output voltage (e.g., during power down). Current is conducted from the output to the input and is not internally limited. If extended reverse voltage operation is anticipated, external limiting might be appropriate.

The TPS715xx features internal current limiting. During normal operation, the TPS715xx limits output current to approximately 500 mA. When current limiting engages, the output voltage scales back linearly until the overcurrent condition ends. Take care not to exceed the power dissipation ratings of the package.

Programming the TPS71501 Adjustable LDO Regulator

The output voltage of the TPS71501 adjustable regulator is programmed using an external resistor divider as shown in Figure 14. The output voltage operating range is 1.2 V to 15 V, and is calculated using:

APPLICATION INFORMATION (continued)

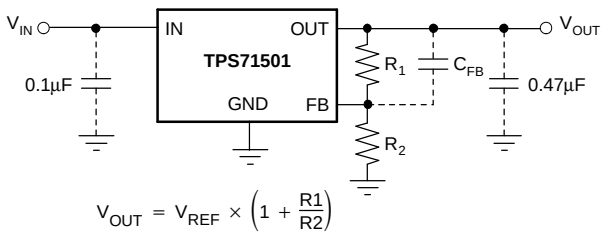
$$V_{OUT} = V_{REF} \times \left(1 + \frac{R1}{R2}\right) \tag{3}$$

where:

- V_{REF} = 1.205 V typ (the internal reference voltage)

Resistors R1 and R2 should be chosen for approximately 1.5-µA divider current. Lower value resistors can be used for improved noise performance, but the solution consumes more power. Higher resistor values should be avoided as leakage current into/out of FB across R1/R2 creates an offset voltage that artificially increases/decreases the feedback voltage and thus erroneously decreases/increases V_O. The recommended design procedure is to choose R2 = 1 MΩ to set the divider current at 1.5 µA, and then calculate R1 using:

$$R1 = \left(\frac{V_{OUT}}{V_{REF}} - 1\right) \times R2 \tag{4}$$



OUTPUT VOLTAGE PROGRAMMING GUIDE

OUTPUT VOLTAGE	R1	R2
1.8 V	0.499 MΩ	1 MΩ
2.8 V	1.33 MΩ	1 MΩ
5.0 V	3.16 MΩ	1 MΩ

Figure 14. TPS71501 Adjustable LDO Regulator Programming

PACKAGING INFORMATION

Orderable Device	Status [1]	Package Type	Package Drawing	Pins	Package Qty	Eco Plan [2]	Lead/Ball Finish	MSL Peak Temp [3]
BQ71525DCKR	ACTIVE	SC70	DCK	5	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM
BQ71533DCKR	ACTIVE	SC70	DCK	5	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM
BQ71533DCKRG4	ACTIVE	SC70	DCK	5	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM
TPS71501DCKR	ACTIVE	SC70	DCK	5	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM
TPS71501DCKRG4	ACTIVE	SC70	DCK	5	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM
TPS71518DCKR	PREVIEW	SC70	DCK	5	3000	TBD	Call TI	Call TI
TPS71519DCKR	PREVIEW	SC70	DCK	5	3000	TBD	Call TI	Call TI
TPS71523DCKR	PREVIEW	SC70	DCK	5	3000	TBD	Call TI	Call TI
TPS71525DCKR	ACTIVE	SC70	DCK	5	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM
TPS71525DCKRG4	ACTIVE	SC70	DCK	5	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM
TPS71530DCKR	ACTIVE	SC70	DCK	5	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM
TPS71530DCKRG4	ACTIVE	SC70	DCK	5	3000	TBD	Call TI	Call TI
TPS71530QDCKRQ1	ACTIVE	SC70	DCK	5	3000	TBD	CU NIPDAU	Level-1-260C-UNLIM
TPS71533DCKR	ACTIVE	SC70	DCK	5	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM
TPS71533DCKRG4	ACTIVE	SC70	DCK	5	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM
TPS715345DCKR	PREVIEW	SC70	DCK	5	3000	TBD	Call TI	Call TI
TPS71550DCKR	ACTIVE	SC70	DCK	5	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM
TPS71550DCKRG4	ACTIVE	SC70	DCK	5	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-1-260C-UNLIM

[1] The marketing status values are defined as follows:
ACTIVE: Product device recommended for new designs.
LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.
NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.
PREVIEW: Device has been announced but is not in production. Samples may or may not be available.
OBSOLETE: TI has discontinued the production of the device.

[2] Eco Plan - The planned eco-friendly classification: Pb-Free (RoHS) or Green (RoHS & no Sb/Br) - please check http://www.ti.com/productcontent for the latest availability information and additional product content details.
TBD: The Pb-Free/Green conversion plan has not been defined.
Pb-Free (RoHS): TI's terms "Lead-Free" or "Pb-Free" mean semiconductor products that are compatible with the current RoHS requirements for all 6 substances, including the requirement that lead not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, TI Pb-Free products are suitable for use in specified lead-free processes.
Green (RoHS & no Sb/Br): TI defines "Green" to mean Pb-Free (RoHS compatible), and free of Bromine (Br) and Antimony (Sb) based flame retardants (Br or Sb do not exceed 0.1% by weight in homogeneous material)

[3] MSL, Peak Temp. -- The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

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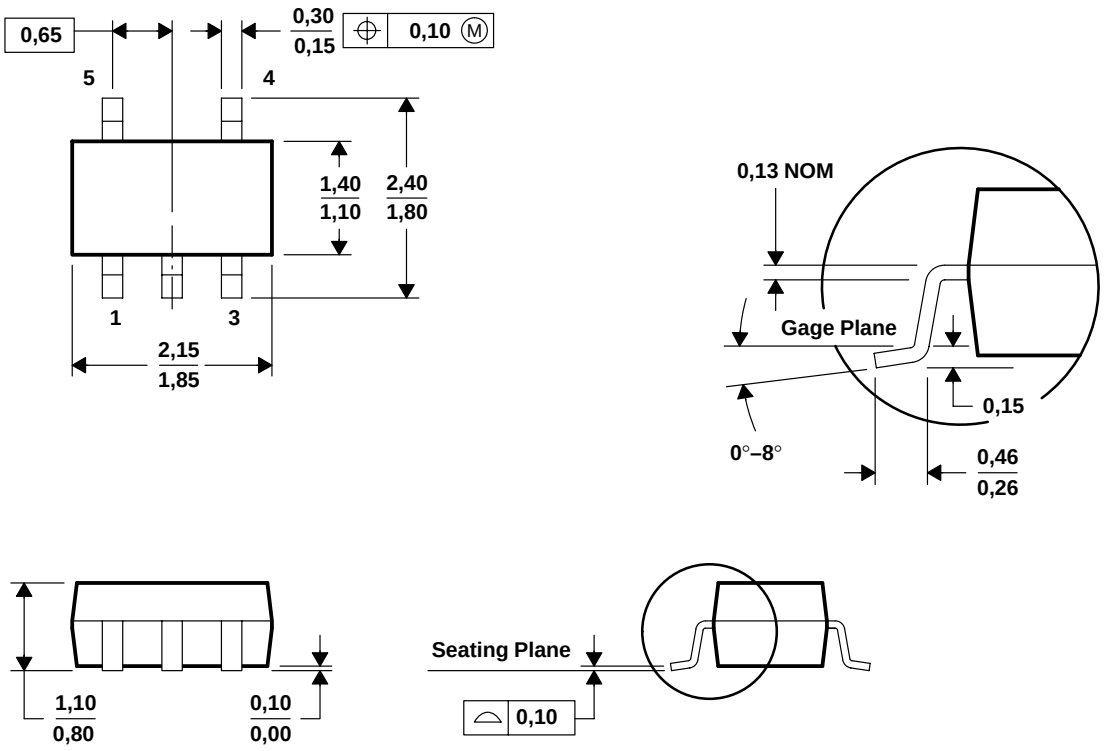
DCK (R-PDSO-G5) **PLASTIC SMALL-OUTLINE PACKAGE**



NOTES: A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusion.
D. Falls within JEDEC MO-203

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